KIRAAT: A Turkish Read-Speech Corpus with Sentence-Aligned Segmentation and a Versioned Selection Policy

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Abstract

We present KIRAAT, a Turkish read-speech corpus comprising 1,840,404 clips and 3,105.7 hours, extracted from 2,680 audiobook and spoken-article recordings across 27 channels. The pipeline transcribes long-form audio, obtains word-level forced alignments, and cuts at sentence boundaries inferred from the transcript. Quality measurements are retained as columns for all generated clips; a separate versioned policy selects a recommended subset of 1,547,494 clips and 2,575.2 hours without deleting the remaining clips from the release.

A comparison on 206 recordings from all 27 channels reports lowercase-start rates of 1.39% for sentence-aligned segmentation and 14.31% for a VAD-based baseline. The corresponding unflagged lowercase-start rates are 0.00% and 12.81%. These transcript proxies do not establish acoustic boundary accuracy. Across the recommended subset, 23 clips have lowercase starts. Exploratory listening audits informed policy revision: all 25 clips in a clipping-ratio audit were judged usable, but only seven exceeded the tested threshold; a word-confidence threshold would exclude 15 of 30 audited clips, 14 of which had correct transcripts.

We report quality distributions, 90 inferred speaker clusters, recording-level training and evaluation partitions with no shared source identifiers or exact normalized transcripts between training and evaluation, and stage throughput on an RTX 3090. Speaker identities and corpus-wide transcription accuracy have not been independently validated, and no downstream TTS experiment is reported. The contribution is a corpus construction and documentation workflow with auditable selection decisions, rather than demonstrated synthesis improvement.

1 Introduction

Read speech is a useful source of paired audio and text for text-to-speech (TTS), but recordings collected for speech recognition do not automatically satisfy synthesis-oriented requirements for sentence boundaries, speaker consistency, text coverage, and acoustic quality. Turkish resources include the recognition-oriented Turkish Speech Corpus [14]. The question here is how to segment and document long-form Turkish readings for potential use in synthesis, rather than whether recognition data exist.

We investigate two questions: how do sentence-based and VAD-based segmentation differ on the same recordings under transcript-based boundary checks, and how can quality selection remain auditable while preserving generated clips? KIRAAT combines an audiobook and spoken-article corpus with per-clip measurements and a versioned selection policy. We evaluate corpus properties and implementation behavior. Transfer to other languages and improvements in TTS quality remain questions for subsequent experiments.

Contributions.

1. **Corpus and documentation.** 1,840,404 clips, 3,105.7 hours, 27 channels and 90 inferred speaker clusters, with per-clip measurements and explicit selection decisions (Section 3).
2. **Segmentation comparison.** Sentence-based and implemented VAD-based strategies are compared on shared recordings and word timestamps, using explicit transcript-boundary proxies (Section 6.1).
3. **Versioned selection.** Measurement stages cannot write recommendation decisions; a policy selects a subset without deleting the remaining generated clips (Section 5).
4. **Exploratory audits.** Small score-stratified listening samples inform threshold revision, with their interpretation restricted to the audited items (Section 6.3).
5. **Long-form implementation.** Windowed forced alignment and a CPU/GPU scheduler support multi-hour recordings; stage throughput is reported (Section 7).

2 Related work

Construction pipelines. Emilia [8, 9] combines standardization, source separation, speaker diarization, VAD segmentation, transcription and quality filtering. GigaSpeech 2 [18] combines automated collection, Whisper-based transcription and alignment with label refinement for speech recognition. WhisperX [2] also separates transcription from forced alignment. Our pipeline uses established components, including Whisper [16] and MMS [15]; we do not claim these components as new methods.

Segmentation. Sentence-oriented cutting is established in LibriTTS [19], Libriheavy [10] and GigaSpeech [3]. WenetSpeech4TTS [13] revises boundaries and addresses speaker mixing when adapting an ASR corpus for synthesis. Our contribution is a comparison on Turkish recordings with shared timestamps and explicit flags for exceptional cuts. It is not a claim that controlled segmentation analysis is absent from the literature.

Selection and restoration. Emilia retains clips with DNSMOS P.835 OVRL above 3.0; its thresholds can be adjusted for different uses [8]. WenetSpeech4TTS releases quality subsets, while LibriTTS-R [11] restores audio while retaining the same constituent samples. We retain generated clips across four partitions and separate their measurements from a versioned recommended-subset policy. This supports alternative selection over stored measurements, subject to screening and missingness. No universal superiority over filtering or restoration is established.

Turkish resources. Publicly documented Turkish speech resources were designed primarily for recognition. Common Voice [1] is a multilingual corpus, including Turkish, whose recordings are both collected and validated by crowdsourcing, and which its authors describe as public domain. FLEURS [4] provides approximately 12 hours per language of read FLoRes-101 sentences under CC BY 4.0, as a multilingual evaluation benchmark. MediaSpeech [12] is a manually transcribed ASR evaluation set collected from official media YouTube channels in Turkish and three other languages. The Turkish Speech Corpus was introduced for multilingual Turkic speech recognition [14]. These resources differ from KIRAAT in recording style and validation: short prompted sentences or broadcast speech with human or crowd verification, as opposed to long-form audiobook and spoken-article readings with automatic transcripts and no human transcript verification. Access also differs: KIRAAT is gated and declares no license

recording length	recordings	hours	share of corpus
< 10 min	168	20.4	0.6%
10–30 min	911	274.0	8.0%
30–60 min	635	462.8	13.5%
1–2 h	467	654.7	19.0%
2–4 h	351	973.4	28.3%
4–8 h	131	700.5	20.4%
8 h and over	35	354.4	10.3%

Table 1: Length distribution of the 2,698 readable source recordings.

(Section 9). We distinguish total transcribed speech from independently validated synthesis training material: neither dataset size nor an ASR label alone establishes TTS suitability.

3 Source material and its measured limits

The recorded inventory comprises 2,700 files from 27 public Turkish audiobook and spoken-article channels: 2,698 readable audio files, one corrupt m4a and one README, totaling 3,440.2 hours and 200.1 GB. The run database contains 2,699 source entries, one per audio file. Of these, 2,680 contribute clips; one (the corrupt m4a) fails decoding, and 18 decode but receive no ASR words and hence no clips (34.8 source hours; 15 of the 18 from one channel). A per-source ledger derived from the run database lists source ID, channel, file name, duration, status and clip count for all 2,699 entries (`docs/source_ledger.csv`). The run database records 3,430.5 hours for the 2,698 decoded files; the 9.7-hour difference from the inventory total reflects separately measured durations and is not reconciled file by file.

Recording duration. The median recording lasts 41.4 minutes and the longest 14.92 hours. Table 1 reports the inventory distribution. About 30.7% of source hours occur in recordings longer than four hours. Libriheavy aligns long-form audio to available book text; we instead align ASR transcripts, because reference book texts were not supplied with these recordings.

Encoding and bandwidth. The inventory reports 2,618 AAC files (3,379.7 hours) and 80 MP3 files (60.5 hours), with 2,693 files at 44.1 kHz. Bit-rate percentiles are 128/129/130 kb/s (p05/p50/p95). Bandwidth was probed on one 60-second excerpt per channel using a 32k-point FFT. The reported median cutoff is 15.7 kHz (range 13.0–15.8); median energy above 12 kHz is 0.05%. These 27 excerpts inform the choice of a 24 kHz release rate, but do not establish bandwidth for every recording or the optimal sampling rate for TTS.

Concentration. The largest two channels contribute approximately 35% of raw hours. No per-channel cap is applied. Each clip retains channel and duration metadata so consumers can study or rebalance concentration. In the recommended subset, the two largest channels account for 38.2% of hours.

4 Pipeline and measurements

The pipeline uses a SQLite state store and a scheduler overlapping CPU and GPU work with one database writer. Table 2 lists configured stages. Quality selection is applied after measurement rather than inside clip measurement stages.

stage	what it does	model / setting
`prepare`	decode, mono, up to 24 kHz, 40 Hz high-pass; no limiter	ffmpeg
`asr`	long-form transcription with word timestamps	faster-whisper `large-v3`, rev. `edaa852e`, beam 5, temperature 0, `condition_on_previous_text: false`
`align`	forced alignment, per-word score	torchaudio `MMS_FA` (with `<star>`)
`boilerplate`	channel-level mining of station idents and announcements	word n-grams, threshold 40% of a channel's recordings (min. 2), `max_words: 24`
`segment`	cut at sentence boundaries, audio-based boundary refinement	`kiraat/segment.py`, `kiraat/boundaries.py`
`clip_qc`	speech ratio, internal silence, edge silences, level, LUFS	silero VAD; BS.1770
`music`	music-to-speech level and AudioSet music score	torchaudio `HDEMUCS_HIGH_MUSDB_PLUS`; `MIT/ast-finetuned-audioset-10-10-0.4593`
`dnsmos`	P.835 SIG / BAK / OVRL	DNS-Challenge `sig_bak_ovr.onnx`, sha256 pinned in config
`speaker`	recording embeddings and within-recording consistency	ECAPA-TDNN `speechbrain/spkrec-ecapa-voxceleb`, revision configured but not passed to loader
`export`	manifest and run record	`manifests/clips.jsonl`, `run.json`

Table 2: Stages and configured settings. The export code records git state, stage versions, configuration and model identifiers; speaker-weight provenance is incomplete.

4.1 Sentence-oriented segmentation

ASR runs over speech windows and outputs words and timestamps. MMS forced alignment refines word times; the transcript is divided into sentences, and audio cuts follow their word spans. VAD supplies ASR regions and clip quality measurements, but does not define the primary sentence boundaries.

The configuration uses a 1.5 s minimum clip duration, a 7.0 s joining target, a 15.0 s ceiling, a 1.2 s maximum joining gap, and leading/trailing padding of 0.15/0.25 s. Exceptional intra-sentence cuts receive `forced_split` or `gap_split` flags. Boilerplate mining uses repeated channel-level word n-grams, separately marking announcements. These flags are construction annotations, not human judgments of boundary correctness.

The aligner forms approximately 30 s windows, with a 60 s maximum and 0.5 s padding, using coarse ASR timestamps. A window reader loads and resamples only the required interval, bounding the alignment waveform and lattice size. This does not establish constant memory use for all stages.

4.2 Acoustic boundary refinement

An energy-based search near an inter-clip word boundary adjusts the cut within a configured window to reduce phoneme truncation. The recorded post-refinement listening audits found no truncated syllables in 28 clips and no truncated words in another 30 clips (20 suspected cases and 10 controls, shuffled). These selected samples do not establish a corpus-wide acoustic error rate. Listener counts, expertise, clip identifiers, annotation instructions and per-item labels are not part of this release, so these audits cannot be independently re-scored.

4.3 Duplicate heuristic

Separate readings of the same text are retained for prosodic diversity. The implemented heuristic groups clips by Turkish-aware normalized text and exact equality of duration, LUFS, RMS and peak; channel is not in the key. Matching summary statistics do not prove waveform identity, and re-encoding or shifted boundaries can prevent true duplicates from matching. The heuristic flags 576 clips. Clips with missing fingerprint fields bypass the check. Its precision and recall have not been independently established.

4.4 Audio preparation and quality columns

Preparation converts audio to mono, applies a 40 Hz high-pass filter and resamples to at most 24 kHz, without upsampling lower-rate sources at this stage. Export declares 24 kHz audio for dataset loading. These operations modify the waveform; no loudness normalization or peak limiting is applied. Source separation is used for measurement, and its output is discarded.

Clip measurements include VAD speech ratio and silence, peak/RMS/LUFS, clipping ratio, DNSMOS P.835 SIG/BAK/OVRL [17], word scores, and speaker embeddings. DNSMOS is a model-derived quality estimate, not a human MOS collected for this corpus. Acoustic properties and quality scores are retained even when the selection policy rejects a clip.

Music screening. The AST AudioSet model [7] scores music labels for every clip. HDemucs [5] runs only when that score is at least 0.05. Separation was skipped for 1,455,201 clips and performed for 385,203 clips in the stored run. Skipped clips receive the sentinel −80 dB in `music_to_speech_db`, and `music_db_separated` is false. This sentinel does not demonstrate absent music. For separated clips the column estimates accompaniment-to-vocal energy, not calibrated speech/noise level or perceptual audibility. A policy requiring separation below the screening threshold needs further model inference.

Text fields and confidence. `text_raw` joins ASR words; `text` applies light cleaning; `text_spoken` expands numerals and abbreviations. Per-operation transformation logs are not exported. The last field is intended for TTS training, but normalization has not been validated corpus-wide against human transcripts. `word_confidence` is the minimum word score, using ASR scores in the current configuration; forced-alignment mean/minimum scores are separate. Neither is a calibrated probability of transcript correctness.

5 The recommended subset policy

The stage base class rejects clip-stage outputs containing `recommended` or `decision`. Operational thresholds are specified in configuration; implementation helpers also contain defaults. The configuration loader rejects unknown sections and keys. The column table is generated from the schema, while narrative documentation and remote cards require separate checks.

Version 6 requires `speech_ratio` ≥ 0.60, `internal_silence_sec` ≤ 1.0, and absence of `oversize`, `forced_split`, `gap_split`, `short`, `duplicate`, `boilerplate` and `background_music` flags. The result is 1,547,494 clips and 2,575.2 hours: 84.1% of generated clips and 82.9% of their duration. Exclusion reasons and policy version are retained. Missing required policy measurements fail the applicable rule.

All generated clips remain in the release. This guarantee begins after segmentation; it does not mean that every source interval or failed decoding result becomes a clip. Alternative selection is possible over stored measurements, but missing or screened values limit policies that can be recomputed without inference.

measure	sentence-aligned	silence-aligned
clips	141,379	76,466
hours	240.54	254.18
lowercase start	1,967 (1.39%)	10,943 (**14.31%**)
no final punctuation	2,405 (1.70%)	12,117 (15.85%)
unflagged lowercase start	**0 (0.00%)**	9,798 (**12.81%**)
unflagged missing final punctuation	0.01%	11.59%
both transcript proxies	418	2,673
median duration	5.94 s	12.90 s
duration, 5th–95th pct	2.69–10.96 s	5.10–15.40 s
at the duration ceiling	0.64%	14.70%
mean `align_score`	0.9442	0.9459
mean of per-clip *minimum* `align_score`	**0.7319**	0.6469

Table 3: Reported comparison on 206 recordings from all 27 channels. Hours are sums of clip durations, including padding, not unique source coverage. Boundary checks use transcript proxies. The baseline is the implemented VAD strategy, not an execution of another published pipeline.

6 Results

6.1 Segmentation comparison

The research notebook reports a comparison on 206 recordings from all 27 channels. The sentence arm is read from the full-run database; the VAD arm is constructed from speech regions on the same recordings and shared word timestamps, using the same configuration object. Source selection uses seed 20260906 and up to eight recordings per channel. Coverage of all channels does not imply equal audio duration per channel.

The arms also differ in grouping, exceptional-cut flags, boilerplate handling, and boundary refinement. In particular, the VAD implementation does not use the sentence joining target in the same way. This is a strategy comparison, not an isolated intervention on a single cut point or a benchmark against executed Emilia or LibriTTS pipelines.

We define *lowercase start* using the Turkish-aware `is_lower_start` routine (the first alphabetic character after leading punctuation is tested; an initial numeral is not flagged). A second proxy tests absence of sentence-final punctuation after stripping closing quotes/brackets. These proxies depend on ASR casing and punctuation, which also guide sentence segmentation. Consequently they are not independent labels of acoustic truncation or sentence integrity.

The reported lowercase-start rate is 1.39% for sentence segmentation and 14.31% for the implemented VAD baseline. Among all clips, unflagged lowercase-start rates are 0.00% and 12.81%, respectively. "Unflagged" means an empty flag list, although the arms do not have identical flag vocabularies. The metadata lack a dedicated flag for many proxy-positive baseline clips; their text still permits the proxy to be computed. They should therefore not be described as undetectable.

The sentence arm sums to 240.54 clip-hours and the VAD arm to 254.18. These are padded clip-duration totals, not measured unique source coverage; VAD padding can overlap between adjacent clips. Retained speech coverage, overlap and omitted words must be reported before interpreting the difference as yield. The values are taken from the research notebook. The per-source outputs of the baseline arm are not part of this release, so the comparison cannot be re-derived from the released database alone; the implementation is `scripts/ablate_segmentation.py`.

Mean alignment scores are 0.9442 and 0.9459. Mean per-clip minimum scores are 0.7319 and 0.6469, but the arms have different word-count and duration distributions. Longer clips have more opportunities for a low minimum, so this statistic alone does not establish poorer

subset	clips	hours	lowercase start	no final punctuation
whole corpus	1,840,404	3,105.7	27,483 (1.49%)	33,346 (1.81%)
recommended subset	1,547,494	2,575.2	**23 (0.0015%)**	430 (0.03%)
outside the policy	292,910	530.5	27,460 (9.37%)	32,916 (11.24%)

Table 4: Transcript-boundary proxies over the full corpus. Rates are per clip; these are not human-labeled acoustic error rates.

edge alignment. No paired recording-level confidence interval or significance test is reported.

6.2 Transcript-boundary proxies over the corpus

The same routines applied to all generated clips yield the counts in Table 4. The recommended subset contains 23 lowercase starts and 430 clips without final punctuation; the former is approximately 15 per million clips, a descriptive proxy rate rather than an acoustic failure probability.

Of the 27,483 lowercase-start clips, 97.6% carry `forced_split` or `gap_split`. Among 1,558,409 clips with no flags, 23 have lowercase starts. The flags make exceptional construction paths explicit; they do not validate the remaining clips. The notebook associates the remaining lowercase-start cases with boilerplate handling, which requires item-level inspection before a causal conclusion.

6.3 Exploratory listening audits

The documented procedure hides channel and measurement values, samples across score bands, shuffles clips and records judgments before revealing scores. The notebook describes six policy/refinement rounds: two on music, two on boundaries, one on clipping and one on word confidence. The exact listener count, inter-rater agreement and per-item labels have not been archived with the available source materials. We therefore treat these as exploratory audits, not population estimates of error or training utility.

Clipping ratio. A candidate ceiling of 0.002 excluded 536 clips in a development sample; 532 were from one channel. These are not full-release exclusion counts. Among 25 audited clips across six score bands (0–0.0227), 24 were judged free of audible distortion and one slightly distorted; all were judged usable. Only seven exceeded the ceiling, and none of those seven was judged audibly distorted. This motivated removal of the candidate rule in policy v5. It does not establish that clipping is harmless in general.

ASR word confidence. Thirty audited clips span six bands from 0.18 to 0.99. Twenty-nine transcripts were judged verbatim correct and one had a small error. A minimum confidence of 0.60 would exclude 15 clips, 14 of which had correct transcripts. The rule was removed in v6. The reported rank correlation with alignment score is +0.262, but agreement between two automatic scores is not independent validation of transcript accuracy. This sample cannot supply corpus WER/CER.

Music. An initial audit found clearly audible music in 6/6 clips in the −40 to −33 dB band. A later 36-clip audit, sampled across four bands, reported audible music in 86% of flagged clips and no music in 11/14 unflagged clips. These stratified observations are not prevalence-weighted precision or recall. The current flag requires both separated level above −40 dB and AudioSet score ≥ 0.3. The notebook reports Spearman +0.479 between the scores on 3,097 separated clips; this screened sample does not characterize clips for which separation was skipped.

column	p05	p50	p95	mean
duration (s)	2.76	5.82	10.56	5.99
`loudness_lufs`	−27.99	−20.92	−12.54	−20.61
`dnsmos_ovrl`	2.862	3.296	3.491	3.253
`dnsmos_sig`	3.271	3.563	3.710	3.536
`dnsmos_bak`	3.631	4.109	4.210	4.041
`speech_ratio`	0.851	0.977	1.000	0.955
`music_score_audioset`	0.001	0.006	0.228	0.034
`music_to_speech_db`	−80.0	−80.0	−40.94	−75.42
`align_score_mean`	0.858	0.961	0.994	0.946
`align_score_min`	0.278	0.800	0.976	0.740
`word_confidence`	0.550	0.917	0.999	0.866
`n_words`	5	11	21	11.7
`internal_silence_sec`	0.000	0.000	0.664	0.166

Table 5: Column distributions over the recommended subset (1,547,494 clips). The music dB column includes an unmeasured sentinel; its summary is not the distribution of measured music levels.

DNSMOS. No DNSMOS threshold is used in the policy. Across the whole corpus, 83.57% of clips have OVRL ≥ 3.0. This describes the effect of one hypothetical threshold, not its validity for Turkish synthesis data.

6.4 Quality distributions

Table 5 describes the recommended subset. Across the whole corpus, mean DNSMOS OVRL is 3.208 (p05/p50/p95: 2.69/3.27/3.49). These are predicted scores. The loudness spread remains in the released audio. The music dB summaries mix measurements with sentinel values and must be interpreted with `music_db_separated`.

Whole-corpus flag counts are `background_music` 207,196, `forced_split` 52,501, `short` 20,904, `oversize` 6,731, `gap_split` 5,141, `boilerplate` 1,053 and `duplicate` 576. Flags overlap, so their counts cannot be added as disjoint exclusions.

6.5 Inferred speaker clusters

ECAPA-TDNN [6] embeds up to 12 deterministically selected clips of at least 3 s per recording. The stage exports a normalized centroid and within-recording clip-pair cosine statistics. Recordings with fewer than two eligible clips do not receive a centroid; this is not speaker diarization.

Average-linkage clustering uses a similarity threshold derived from two proxy distributions: per-recording within-clip mean similarity, and centroid similarity between different channels (up to 200,000 sampled pairs, seed 20260907). The code equalizes their empirical tail rates. These proxies are not labeled same-speaker and different-speaker pairs: one recording can contain multiple voices, and one narrator can occur on different channels. They also aggregate different types of similarity observations.

The notebook reports threshold 0.532, proxy tail error 0.52%, and 90 clusters over 2,677 recordings with embeddings; 21 readable recordings lack an embedding. The 0.52% is not a speaker-verification EER measured against known identities. Median cluster margin is reported as 0.445; centroids include the recording itself, so positive margins are not independent validation. The resulting cluster label is released per clip as `speaker_id`, but the recording embeddings are not part of this release, so the clustering cannot be recomputed from released files. No labeled same/different-speaker pairs have been audited.

split	clips	hours	recordings	channels
train	1,512,448	2,517.08	2,579	27
dev	5,750	9.54	40	27
test	5,473	9.30	45	27
rest	316,733	569.78	—	27
total	**1,840,404**	**3,105.70**	**2,680**	**27**

Table 6: The four published splits. `train`, `dev` and `test` are drawn from the recommended subset; `rest` carries every remaining clip, so the four splits together are the whole corpus. The recordings column counts recordings contributing clips to that split; `rest` draws clips from recordings across all three, and 16 of the 2,680 recordings have all of their clips in it.

The reported assignment has more than one cluster in 13 channels and nine clusters spanning several channels. These are inferred relationships, not verified narrator identities. The largest cluster contains 18.2% of hours, the top five 54.4%, and the top ten 75.0%; 17 clusters have less than one hour. Speaker-ID metadata should be treated as provisional clustering labels.

6.6 Evaluation partitions

Recommended clips are assigned to train/dev/test by source recording. Dev and test each target 10 hours, allocated equally across channels, with seed `kiraat-2026`. Equal targets do not guarantee equal final channel shares after transcript-overlap removal and quota trimming. No human-verified test transcript set is currently supplied.

The remaining 316,733 clips are in `rest`: 292,910 non-recommended clips and 23,823 recommended clips unused after quota trimming or text cleanup. Thus recordings can appear in `rest` and one of train/dev/test. Only the three named training/evaluation partitions have disjoint source IDs.

Before cleanup, 2,263 test clips and 2,723 dev clips shared normalized text with training. Evaluation clips with these keys are moved to `rest`. The normalization applies Unicode NFC, Turkish-aware lowercasing, punctuation removal and whitespace collapsing. After cleanup, no train–dev or train–test key overlaps remain. This does not exclude near-duplicate text, shared books, re-uploads under distinct source IDs or pretraining contamination. A separate database audit finds 12 shared normalized text keys between dev and test. These pairs do not share source IDs, but their text is not independent.

The reported cluster counts are 89 in train and 26 each in dev/test, with 25 of 26 test clusters also present in train. This is predominantly a seen-cluster evaluation, subject to cluster-label uncertainty. Consumers expanding training with `rest` must exclude every dev/test source ID and recheck text overlap. Changing the selection policy does not preserve these partition guarantees automatically.

7 Throughput and reproducibility

The notebook reports processing on an RTX 3090 (24 GB), with 12 CPU workers and a 280 W GPU power cap. Reported stage rates are 94 audio-hours per hour for alignment, 46 clips/s for QC, 28 clips/s for music (two GPU workers on this machine), 81.1 clips/s for DNSMOS, and approximately one second per recording for speaker embeddings. They are stage measurements, not an end-to-end runtime or measured energy budget. CPU model, host-memory peak, exact profiling inputs and total wall time remain undocumented here. The notebook reports approximately 567 GB of working storage, including 284 GB of prepared audio and 246 GB of clips; these intermediate files are not included in the metric-database archive.

The export implementation records git state, stage versions, configuration, package versions and model identifiers in `run.json`. However, the speaker loader does not pass the configured revision, and the provenance model/package lists omit the speaker model, SpeechBrain and scikit-learn. Consequently complete weight pinning cannot be claimed. The original run record, per-source ablation outputs, speaker embeddings and listening labels are not part of this release; the source ledger is. A commit reference alone is insufficient when the run used uncommitted changes.

The stored full-run database can be checked with `scripts/verify_card.py`. The 150 implemented checks pass on the stored database, including regenerated duplicate flags, policy decisions and partition counts. This verifies consistency of stored results with the card, not waveform integrity, human transcript accuracy, speaker labels or the missing baseline outputs. Models and the full pipeline were not rerun.

8 Limitations

No downstream synthesis evaluation. No TTS model trained on this corpus is evaluated. Claims concern construction and documentation, not improved intelligibility, naturalness or voice similarity. An equal-budget comparison of corpus selections would test utility in a future model study; it is not needed to report a clearly scoped corpus.

Automatic transcripts and alignments. WER/CER against an independently transcribed representative sample is absent. Casing/punctuation checks depend on the same ASR output used for segmentation. Numerals fall back to ASR timing when the aligner cannot represent them; a separate spoken-form mapping could preserve the published text, but has not been evaluated. Short sentence-initial words can receive implausible alignments. The notebook's affected-word/clip counts have unclear denominators and are not used as a prevalence estimate here.

Audit selection and adaptive thresholds. Small score-stratified audits guided threshold choice on development samples. Listener multiplicity and inter-rater reliability are undocumented; independent held-out validation is required to estimate policy error. Zero observed errors in selected samples do not establish zero population error.

Concentration and coverage. Channel and cluster shares are highly unequal. Genre, accent, demographic and lexical coverage have not been systematically characterized. Most source files are similarly encoded AAC, with a smaller MP3 component. Boilerplate mining misses variable-slot templates. Speaker labels and duplicate heuristics are unvalidated, and dev/test transcripts are not manually verified.

9 Ethics, rights and access

The material was collected from public YouTube audiobook and spoken-article uploads. Public accessibility alone does not document permission to redistribute the underlying audio or literary texts. The public dataset page declares no dataset license and states that recording rights remain with their holders; this manuscript consequently does not claim a license for third-party recordings or unrestricted commercial reuse. Source-by-source redistribution rights and narrator consent have not been documented here, so unrestricted reuse cannot be established.

Voices can identify narrators even when cluster IDs are pseudonymous. The transcripts have not been manually screened for personal or sensitive content. The public card offers a takedown route through repository issues or dataset discussions. The paper does not establish

that takedown substitutes for permission or that voice-cloning uses have been authorized. A final resource statement should distinguish recording rights, annotation/code licensing, access conditions, removal procedure and intended or restricted uses.

10 Conclusion

KIRAAT documents 3,105.7 hours of Turkish read speech and a versioned policy selecting 2,575.2 hours while retaining all generated clips. The reported segmentation comparison shows lower transcript-proxy violation rates for the sentence-based strategy than for the implemented VAD baseline. Exploratory listening informed removal of two candidate thresholds. These observations support auditable data construction; they do not establish acoustic error rates, verified speaker identities or better TTS output. Human reference annotations, archived experimental artifacts and resolved rights/access statements remain necessary to strengthen the resource release.

Availability

The public dataset landing page is `https://huggingface.co/datasets/serdarcaglar/kiraat`; file access requires logging in and accepting the page's conditions. The numbers in this paper refer to dataset revision `75b3297b0d9d3bb00547c7c4ab871f53f5a0fc79` (733 Parquet shards). The pipeline repository is `https://github.com/serdarildercaglar/kiraat`. The metric-database release is linked at `https://github.com/serdarildercaglar/kiraat/releases/tag/db-2026-09-07`. `docs/release_checksums.sha256` in the repository lists the SHA-256 of every shard of that revision and of the database archive, as reported by the hosting services; `docs/source_ledger.csv` is the source ledger. The database checks do not independently validate the remote audio shards. No dataset license is declared, and no repository LICENSE file is included in the current source tree.

Editorial assistance. OpenAI Codex assisted with language revision and consistency checks against the implementation and stored metrics. This assistance did not include training a TTS model or collecting new human annotations. The author remains responsible for verifying the manuscript and its references before submission.

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